

MUHAMMAD WAIZ IMRAN

Cloud & Data Engineer

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PROFESSIONAL SUMMARY

Multi-Cloud Data Engineer with hands-on experience across AWS and Azure, building real-time streaming pipelines (Kafka, Spark, Flink) and lakehouse/warehousing solutions (Databricks, Delta Lake, Snowflake, dbt). Proven ability to architect end-to-end ETL/ELT systems and secure, governed data infrastructure at enterprise scale. Seeking a Data Engineer role to deliver scalable, high-impact data infrastructure globally.

EXPERIENCE

Cloud & Data Engineering Apprenticeship | SMIT (Saylani Mass IT Training) Jun 2024 – Jun 2026 · Karachi, PK

- Completed a 2year structured Cloud Data Engineering Apprenticeship covering end to end data pipeline design, cloud infrastructure, big data processing, and modern data stack tooling.
- Built production grade AWS pipelines spanning IoT Core, Lambda, DMS, Glue (PySpark), S3, Athena, Redshift, and QuickSight debugging 8+ real infrastructure blockers including VPC, CDC privileges, and Glue schema mismatches.
- Architected realtime streaming systems using Apache Kafka, Apache Spark, and Apache Flink; developed Snowflake Medallion Architecture with dbt transformations and Airflow orchestration.

Cloud & Data Engineer Intern | Zaitoon Ashraf Jun 26, 2026 – Present · Karachi, PK

- Applying cloud data engineering skills in a live production environment contributing to AWSbased data pipeline development, ETL workflow optimization, and cloud infrastructure management.
- Collaborating with the engineering team on realworld data integration challenges, supporting data ingestion, transformation, and analytics delivery using AWS services and modern data stack tools.

KEY PROJECTS

EndtoEnd IoT Data Pipeline on AWS | IoT Core · Lambda · EC2 · RDS · DMS · Glue (PySpark) · S3 · Athena · QuickSight · CloudFormation

- Designed and deployed a 12+ services, production grade IoT data pipeline on AWS that ingests realtime temperature and humidity sensor data, migrates it through a simulated onpremise database to the cloud, applies Spark based transformations, and delivers BI analytics.
- Simulated 5 virtual IoT sensors via CloudFormation publishing MQTT readings every 5 seconds to IoT Core; built a Python Lambda function writing to MariaDB on a private subnet EC2 instance with credentials managed in AWS Secrets Manager; configured Kinesis Firehose as a parallel S3 backup stream.
- Authored a Glue 4.0 PySpark ETL job with manual schema definition, nulldropping, and Parquet output reducing Athena query cost by ~80% and built QuickSight dashboards surfacing temperature trends, humidity metrics, and device alerts across 1,069+ sensor rows processed.

Enterprise Fraud Detection Lakehouse — Azure Databricks | Azure Databricks · Delta Lake · Azure Event Hub · Apache Spark · Unity Catalog · MLflow · Azure Key Vault · GitHub Actions · Power BI

- Studied FINRA's publicly documented fraud-surveillance case study, identified architectural gaps in latency, governance, and scalability, and independently redesigned it as a real-time Medallion Lakehouse on Azure Databricks (Bronze/Silver/Gold via Delta Lake).
- Designed a real-time ingestion layer with Azure Event Hub/Apache Kafka feeding Databricks Autoloader into the Bronze layer, with Apache Spark Structured Streaming handling cleansing, validation, and aggregation through Silver and Gold layers.
- Engineered a feature engineering and ML pipeline (trade velocity, risk features, market indicators) powering XGBoost, Isolation Forest, and AutoEncoder models for anomaly and fraud scoring — tracked via MLflow Registry and served through Databricks Model Serving.
- Implemented an enterprise security and governance layer — Azure Entra ID, Unity Catalog, Azure Key Vault, RBAC, and encryption — alongside GitHub Actions CI/CD, Lakehouse/Azure Monitor observability, and a fraud-analytics business layer with Power BI dashboards, Teams/email alerting, and an investigation-portal integration.

AWS DevOps CI/CD Pipeline | GitHub Actions (OIDC) · ALB · Auto Scaling Group · Multi-AZ RDS PostgreSQL · KMS · Secrets Manager · NAT Gateway · Next.js

- Built a highly available AWS architecture for a Next.js portfolio site, provisioning a MultiAZ RDS PostgreSQL database, Auto Scaling Group, and Application Load Balancer across zones.
- Automated a secure CI/CD pipeline with GitHub Actions using OIDC federation for keyless AWS authentication, deploying builds automatically on every push to production.
- Hardened the environment with KMS encryption, Secrets Manager for credentials, and a NAT Gateway for private subnet access, running on Ubuntu 22.04 EC2 instances.

TECHNICAL SKILLS

Languages: Python, SQL

Cloud Platforms: AWS, AZURE, GCP

Streaming & Big Data: Apache Kafka, Apache Spark (PySpark), Apache Flink, Apache NiFi, Apache Airflow

DevOps & Tools: Docker, Git, Linux, FastAPI, Flask

Visualization: Power BI, Tableau, Looker Studio, Amazon QuickSight

EDUCATION & CERTIFICATIONS

Cloud & Data Engineering SMIT (Saylani Mass IT Training)

Jun 2025 – Jun 2026 · Karachi, PK Karachi, PK

Microsoft Certified : Fabric Data Engineer Associate (DP-700)

Microsoft Certified : Azure Solution Architect (AZ-350)